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ORIGINAL ARTICLE

Influence of priming exercise on muscle deoxy[Hb + Mb] during ramp cycle exercise

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Abstract The aim of the present study was to gain better insight into the mechanisms underpinning the sigmoid pattern of deoxy[Hb + Mb] during incremental exercise by assessing the changes in the profile following prior high-intensity exercise. Ten physically active students performed two incremental ramp (25 W min^{-1}) exercises (AL and LL, respectively) preceded on one occasion by incremental arm (10 W min^{-1}) and on another occasion by incremental leg exercise (25 W min^{-1}), which served as the reference test (RT). Deoxy[Hb + Mb] was measured by means of near-infrared spectroscopy and surface EMG was recorded at the Vastus Lateralis throughout the exercises. Deoxy[Hb + Mb], integrated EMG and Median Power Frequency (MdPF) were expressed as a function of work rate (W) and compared between the exercises. During RT and AL deoxy[Hb + Mb] followed a sigmoid increase as a function of work rate. However, during LL deoxy[Hb + Mb] increased immediately from the onset of the ramp exercise and thus no longer followed a sigmoid pattern. This different pattern in deoxy[Hb + Mb] was accompanied by a steeper slope of the iEMG/*W*-relationship below the GET (LL: 0.89 ± 0.11% W^{-1}; RT: 0.74 ± 0.08% W^{-1}; AL: 0.72 ± 0.10% W^{-1}) and a more pronounced decrease in MdPF in LL (17.2 ± 4.5%) compared to RT (5.0 ± 2.1%) and AL (3.9 ± 3.2%). It was observed that the sigmoid pattern of deoxy[Hb + Mb] was disturbed when the ramp exercise was preceded by priming leg exercise. Since the differences in deoxy[Hb + Mb] were accompanied by differences in EMG it can be suggested that muscle fibre recruitment is an important underlying mechanism for the pattern of deoxy[Hb + Mb] during ramp exercise.

Keywords NIRS · Sigmoid pattern · Muscle fibre recruitment · Bohr effect

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Introduction

Near-infrared spectroscopy is a technique that allows non-invasive registration of muscle oxygenation and hemodynamics. The concentration of deoxygenated haemoglobin and myoglobin (deoxy[Hb + Mb]) can be considered as a valuable parameter since it can provide information on the muscle blood flow to oxygen uptake relationship ($\dot{Q}_m/\dot{V}O_{2m}$-relationship) during dynamic exercise (Ferreira et al. 2007). This relationship is an important determinant of the aerobic exercise capacity in both healthy subjects and patients. Since deoxy[Hb + Mb] has been proposed to be proportional to fractional O_2 extraction and thus to arteriovenous O_2 difference [$C_{(a\text{-}v)}O_2$], it can be suggested that the pattern of deoxy[Hb + Mb] follows a hyperbolic profile as a function of work rate due to the widely demonstrated linear $\dot{Q}_m/\dot{V}O_{2m}$-relationship with a positive intercept on the $\dot{Q}_m$-axis (Delorey et al. 2003; Grassi et al. 2003; Ferreira et al. 2005). Several studies, however,

observed that a sigmoid model provides a better fit to the deoxy[Hb + Mb] response to incremental exercise, both in non-steady state (Ferreira et al. 2007; Boone et al. 2009; Boone et al. 2010; DiMenna et al. 2010) as well as in steady state conditions (Grassi et al. 2003; Boone et al. 2010). This sigmoid pattern indicates that $\dot{Q}_m$ initially increases at a faster rate compared to $\dot{V}O_{2m}$, but then slows down in the low-to-moderate intensity domain to finally increase at a similar rate as $\dot{V}O_{2m}$ as peak $\dot{V}O_2$ is approached. Until now the mechanisms underpinning this sigmoid pattern of deoxy[Hb + Mb] and thus non-linear $\dot{Q}_m/\dot{V}O_{2m}$-relationship have not been established. Two main mechanisms have been proposed to induce a sigmoidal increase in deoxy[Hb + Mb] response as a function of work rate: (a) the Bohr effect, due to the appearance of a lactic acidosis as work rate increases during incremental exercise, induces a rightward shift in the O_2Hb-dissociation curve with a progressive faster increase in deoxy[Hb + Mb] as a consequence (Belardinelli et al. 1995; Bhambhani et al. 1997; Grassi et al. 2003); (b) the sequential recruitment of muscle fibres from predominantly slow-twitch fibres to mixed slow- and fast-twitch fibres as work rate increases might induce a sigmoid increase in $C_{(a\text{-}v)}O_2$, and thus, deoxy[Hb + Mb] due to different $C_{(a\text{-}v)}O_2$ profiles in the different muscle fibres (Ferreira et al. 2007; Boone et al. 2010).

The purpose of the present study was to gain better insight into the underlying mechanisms of the sigmoid pattern of deoxy[Hb + Mb] response to incremental exercise by means of priming high-intensity exercise of different types, i.e. arm and leg exercise. The Bohr effect will have a systemic effect due to the establishment of a metabolic acidosis (and an increase in core temperature), and will be elicited by both the priming leg and arm exercise. However, during the ramp exercise following the priming leg exercise, it can be suggested that sequential recruitment of motor units consisting of different fibre types (Henneman 1985) will be disturbed as a consequence of muscle fatigue from the previous high-intensity leg exercise (Burnley et al. 2002; Krustrup et al. 2004; Layec et al. 2009). Priming high-intensity leg exercise has shown to increase the motor unit recruitment in subsequent exercise (Layec et al. 2009) and according to the size principle (Henneman 1985) the additional muscle fibres that will be recruited are likely to be fast-twitch fibres with a different $C_{(a\text{-}v)}O_2$ profile (Ferreira et al. 2006). To gain insight into the impact of muscle fibre recruitment, surface electromyography was recorded throughout the ramp exercise. Differences in the pattern of deoxy[Hb + Mb] during the ramp cycle exercise when preceded by either priming leg or arm exercise might provide insight into the underlying mechanisms of the deoxy[Hb + Mb] response to ramp exercise. It was hypothesised that priming arm exercise would have no impact on the qualitative response of deoxy[Hb + Mb] to the subsequent ramp exercise. In contrast, priming leg exercise would induce an immediate increase in deoxy[Hb + Mb] due to a disturbance of the pattern of muscle fibre type recruitment, i.e. a more pronounced recruitment of fast-twitch fibres from the onset of the subsequent ramp exercise.

Methods

Subjects

A group of ten male physically active students (PA students), mean ± SD age 22 ± 2 years volunteered to take part in this study. The subjects had a mean body mass of 73 ± 5 kg and a mean height of 1.78 ± 0.05 m and they were all active in a wide variety of sport activities. During their first visit to the laboratory skinfold thickness at the site of application of the NIRS probe on the M. Vastus Lateralis was determined by means of a Harpenden skinfold calliper. The subjects had a mean skinfold of 9.0 ± 4.9 mm. All subjects were informed about the protocol and the aim of the study and signed an informed consent approved by the ethical committee of the university hospital. A medical history questionnaire and an examination including rest and exercise ECG were performed before the start of the study. All subjects were declared to be in good health and none of them presented medical contraindications for participation in the study.

Experimental procedure and protocol

The subjects were asked to come to the laboratory on three occasions. On the first day the subjects performed a 25 W min^{-1} ramp exercise test to exhaustion on an electromagnetically braked cycle ergometer (Excalibur Sport, Lode, Groningen, The Netherlands) which was used to familiarise the subjects to the laboratory setting and with the performance of the ramp protocol. On a second test day two 25 W min^{-1} ramp exercise tests to exhaustion were performed on a cycle ergometer separated by 6 min of rest. The first of the two ramp exercises served as the reference test (RT) to determine the deoxy[Hb + Mb] response to a single ramp exercise. The second ramp exercise (LL) was used to address the effect of the priming high-intensity leg exercise on the deoxy[Hb + Mb] response to ramp exercise. On a third test day the 25 W min^{-1} ramp cycle exercise (AL) was preceded by a priming 10 W min^{-1} ramp exercise to exhaustion on an arm ergometer (Lode, Groningen, The Netherlands) (see also Fig. 1). The two priming exercise conditions were performed in a random order on non-consecutive days. The actual ramp increase in

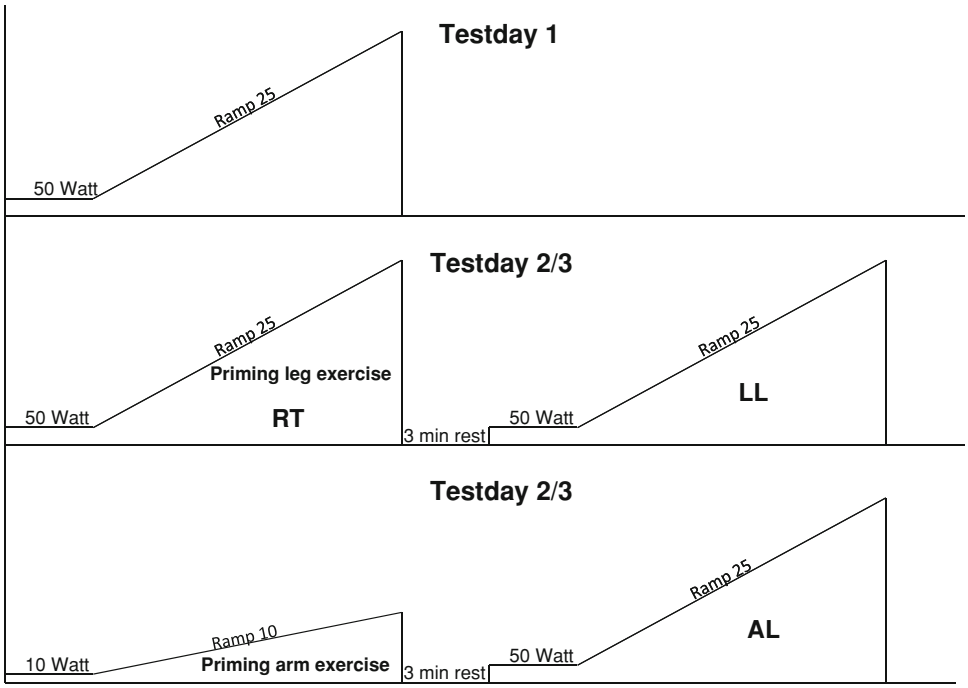

Fig. 1 Schematic representation of the exercise protocol. On the first test day, the subjects performed a ramp 25-protocol to familiarise to the laboratory procedures. The priming leg exercise was used as a reference test (RT). LL was the ramp exercise following the reference RT. AL was the ramp exercise proceeded by a priming arm exercise

work rate in each exercise test was preceded by 3 min of baseline cycling at 50 W for the leg cycling protocols and at 10 W for the arm cycle protocol. In the two conditions with the priming exercises the subjects were allowed to rest on the cycle ergometer for 6 min following the first exhaustive ramp exercise (either the arm or the leg exercise). Each exercise test was terminated when the subjects could no longer maintain the instructed pedal rate of 70 rpm, despite strong verbal encouragement. All exercise tests were performed within a period of 2 weeks to ensure that the level of physical fitness had not changed. The subjects were asked to abstain from strenuous exercise for at least 24 h prior to their visit to the laboratory.

Measurements

Gas exchange

During the exercise tests, $\dot{V}O_2$ was measured continuously on a breath-by-breath basis by means of a computerised O_2–CO_2 analyser-flow meter combination (Jaeger Oxycon Pro, Germany) and averaged over 10 s intervals. Prior to the test, the gas analysers (a paramagnetic O_2-analyser and an infrared CO_2-analyser) were calibrated and a volume calibration ("triple" V transducer) was conducted.

Blood lactate concentration

To assess the degree of metabolic acidosis, blood samples were drawn from the fingertip of middle finger of the right hand immediately following the first exhaustive exercise (either arm or leg cycling), at 2 min of baseline cycling and at 150 W in the second exercise test (LL and AL), to determine blood lactate concentrations (Arkray Inc, Kyoto, Japan).

Near-infrared spectroscopy

Deoxy[Hb + Mb] was measured by means of a frequency-domain multi-distance near-infrared spectroscopy system (Oxiplex TS, ISS, Champaign, IL, USA). The NIRS probe consisted of eight light-emitting diodes, operating at wavelengths of 750 and 830 nm and one detector fibre bundle (source-detector distance = 2.0–3.5 cm). The probe was positioned longitudinally on the distal section of the Musculus Vastus Lateralis of the right leg and secured with Velcro® straps around the thigh. Prior to the placement of the NIRS probe, the NIRS system was calibrated and the skin was carefully shaved. Pen marks were made on the skin to indicate the margins of the probe to check for any downward sliding during the cycling exercise and for accurate repositioning of the probe on the subsequent visit to the laboratory. The deoxy[Hb + Mb] was recorded at a frequency of 25 Hz and afterwards averaged into 1 s values.

Surface EMG

Myoelectric activity was detected by means of surface electromyography (surface EMG) and recorded with bipolar 34 mm diameter Ag–AgCl electrodes (Blue Sensor) at a sampling frequency of 1,000 Hz. The electrodes were placed on the distal half of the M. Vastus Lateralis of

the left leg. Each electrode site was prepared by shaving, abrading and swabbing the site with diluted ethanol. The reference electrode was placed over the spiny process of prominent cervical vertebrae. The EMG signal was checked for movement artefacts and the wires connected to the electrodes were taped on the thigh of the subjects. To make sure that the electrodes would be placed at the same location of the muscle on the next visit to the laboratory, the electrode sites were pen-marked. Myoelectric signals were relayed from the bipolar electrodes to a Telemyo device (Noraxon Inc., Scottsdale, AZ, USA).

Data analysis

Gas exchange

The $\dot{V}O_2$ breath-by-breath values were averaged over 10 s intervals. The gas exchange threshold (GET) of each ramp leg cycle exercise test was determined by means of the V-slope method (Beaver et al. 1986), i.e. the point at which increases disproportionate to $\dot{V}O_2$. The peak $\dot{V}O_2$ was determined as the highest $\dot{V}O_2$ over an interval of 30 s. Both GET and peak $\dot{V}O_2$ were described as a function of body mass. The baseline $\dot{V}O_2$ was calculated as the mean $\dot{V}O_2$ of the final 90 s of the baseline cycling at 50 W. The 10 s $\dot{V}O_2$ values were expressed as a function of work rate and the slope of the $\dot{V}O_2/W$-relationship ($\Delta\dot{V}O_2/\Delta W$) was calculated by means of a linear regression analysis ($y = ax + b$) for the data below (S1) and above (S2) the GET. The first two minutes of the ramp were not taken into account to make sure that the linear increase in $\dot{V}O_2$ had already started. The final minute of the exercise test was also eliminated so that the levelling-off in $\dot{V}O_2$, when the $\dot{V}O_2$max is approached, would not affect the slope of the $\dot{V}O_2/W$-relationship above the GET.

Near-infrared spectroscopy

The 1 s values of the deoxy[Hb + Mb] were normalised to the total amplitude of the response in the RT, LL and AL. Baseline deoxy[Hb + Mb] was calculated as the average of the final 60 s of the 3 min cycling period at 50 W and set to 0%, whereas the peak deoxy[Hb + Mb] was determined as the average of the final 15 s of the exercise test and set to 100%. The normalised deoxy[Hb + Mb] of RT was plotted as a function of work rate and fitted to a sigmoid model to test whether deoxy[Hb + Mb] followed the expected sigmoidal increase. It was expected that the deoxy[Hb + Mb] response would be altered by the priming exercise. Therefore, instead of using the traditional curve fitting techniques to get an insight into the difference in the deoxy[Hb + Mb] response between RT, LL and LL, it was chosen to compare the change in the normalised deoxy[Hb + Mb] was calculated for 30-s bins centred upon 10% increments of the change in work rate from baseline (0%) to end exercise (100%). This method would allow to get a more comprehensive insight into change int the deoxy[Hb + Mb] response across the complete range of work rates between the different exercises.

Surface EMG

The raw EMG signals were rectified, band-pass filtered (5–2,000 Hz) and integrated using commercially available software (MyoResearch2.10, Noraxon Inc., Scottsdale, AZ, USA). The iEMG-values of each ramp exercise test were averaged over 10 s intervals, time aligned to the start of the ramp and averaged. Mirka (1991) reported that normalisation of the EMG data as a percentage of maximal voluntary contraction induces error and therefore, we chose to express the iEMG-values as a percentage of the average values measured at 50 W in each protocol (Mirka 1991). The 5-s iEMG-values were plotted as a function of work rate and the slope of the iEMG/W-relationship (ΔiEMG/ΔW) was calculated for the same data range below the GET as for the $\dot{V}O_2/W$-relationship (see above) by means of linear regression analysis. Above the GET it has been shown that the iEMG increases in a curvilinear fashion and therefore, the ΔiEMG/ΔW above the GET was calculated by only using the values at GET and peakP. The Median Power Frequency (MdPF), as an indication of the distribution of frequency content, was calculated from the raw EMG signal at 0–100% peakP at each 10% increment. The MdPF at baseline cycling in AL en LL were set at 100% and the changes in MdPF were expressed relative to this initial value. This analysis of the frequency content of the EMG signal may reflect changes in muscle fibre type recruitment (Tesch et al. 1983; Gerdle et al. 1991).

Statistical analysis

The mean values ± SD were determined for the important parameters of gas exchange (peak $\dot{V}O_2$, peakP, baseline $\dot{V}O_2$, GET, S1 and S2), NIRS (deoxy[Hb + Mb] at baseline cycling and peakP, the amplitude of the deoxy[Hb + Mb] response and the changes in deoxy[Hb + Mb] at each 10% increment from 0 to 100% peakP) and EMG responses (iEMG at baseline cycling, baseline MdPF and the change in MdPF at each 10% increment from 0 to 100% peakP). These parameters were compared between the RT, AL and LL by means of a One-Way Anova in SPSS 15.0. Any significant main effect was further analysed post hoc by means of a Paired Samples T test. Significance was set at $P < 0.05$.

Results

Gas exchange

In Table 1 the mean values for peak $\dot{V}O_2$, peakP, baseline $\dot{V}O_2$, GET, S1 and S2 are presented. Peak $\dot{V}O_2$, peakP and GET did not differ significantly indicating that the priming exercises did not have an impact on the maximal values or GET of the ramp exercises. However, the baseline $\dot{V}O_2$ in LL was slightly higher (i.e. a trend to significance $P = 0.072$) compared to RT and AL. The slope of the $\dot{V}O_2/W$-relationship below GET (S1) was significantly higher in LL compared to RT and AL, whereas above the GET (S2) it was significantly lower.

Blood lactate concentration

At the end of the priming exercises the blood lactate concentrations were, respectively, 11.40 ± 3.65 and 10.71 ± 4.21 m mol l^{-1} for the LL and AL, respectively. During baseline cycling following the 6 min resting period and at 150 W, the lactate concentrations had slightly decreased to 8.14 ± 2.27 and 7.55 ± 2.12 m mol l^{-1}, respectively, for the LL and to 8.42 ± 3.01 and 7.21 ± 2.88 m mol l^{-1}, respectively, for the AL. The lactate values for LL and AL were not significantly different for any of the three measurements.

NIRS

In Table 2, the absolute concentrations of deoxy[Hb + Mb] at baseline cycling, peakP and the amplitude of the deoxy[Hb + Mb] response for the three conditions are presented. There was no significant difference among the three conditions. The curve fitting procedure revealed that a sigmoid model ($y = -6.1 + 112.0/(1 + \exp^{-(-2.87+0.015x)})$) appropriately ($R^2 = 0.98 \pm 0.03$; SEE = 3.7 ± 3.1) fitted the deoxy[Hb + Mb] response to RT in all subjects. In Fig. 2, the mean deoxy[Hb + Mb] response as a function of %peakP is represented for the three exercise conditions. The change in deoxy[Hb + Mb] during the LL was significantly higher from 10 to 70% peakP compared to RT and AL, whereas RT and AL were not significantly different.

Table 1 Mean values ± SD for peak $\dot{V}O_2$, peak power (peakP), baseline $\dot{V}O_2$, gas exchange threshold (GET) and the slope of the $\dot{V}O_2/W$-relationship below (S1) and above (S2) the GET for the reference test (RT), and the ramp exercises preceded by a priming leg (LL) and arm exercise (AL)

	RT	LL	AL
Peak $\dot{V}O_2$ (ml min^{-1} kg^{-1})	52.1 ± 4.8	51.9 ± 4.9	52.9 ± 4.8
peakP (W)	371 ± 40	344 ± 41	348 ± 42
Baseline$\dot{V}O_2$ (ml min^{-1})	712 ± 83	788 ± 79[c]	720 ± 101
GET (ml min^{-1})	2,660 ± 233	2,735 ± 303	2,702 ± 292
S1 (ml min^{-1} W^{-1})	9.45 ± 0.52	10.04 ± 0.61[a,b]	9.42 ± 0.44
S2 (ml min^{-1} W^{-1})	10.42 ± 0.48	9.87 ± 0.35[a,b]	10.31 ± 0.51

[a] Significantly different from RT

[b] Significantly different from AL ($P < 0.05$)

[c] Trend to significance ($P < 0.10$)

Table 2 Mean values ± SD for baseline, peak and amplitude of the deoxy[Hb + Mb] response for the reference test (RT) and the ramp exercises preceded by a priming leg (LL) and arm exercise (AL)

Deoxy[Hb + Mb]	RT	LL	AL
Baseline (μM)	23.6 ± 8.0	21.7 ± 5.2[a]	22.0 ± 5.9
Peak (μM)	33.2 ± 11.4	32.8 ± 11.5	30.8 ± 9.6
Amplitude (μM)	9.6 ± 4.3	11.1 ± 7.0	8.8 ± 4.9

[a] Trend for significant ($P < 0.10$) difference between LL and RT

Surface EMG

In Table 3 the iEMG at baseline cycling and the ΔiEMG/ΔW below and above GET for RT, LL and AL are presented. In Fig. 3 the iEMG as a function of work rate is displayed for RT, AL and LL in a representative subject. The slope of the iEMG/W-relationship below the GET was significantly steeper in LL compared to RT and AL. Above the GET ΔiEMG/ΔW did not differ significantly between RT, LL and AL, or at peak exercise. In Fig. 4 the mean pattern of MdPF over the work rate range in LL, AL and RT is shown. The initial absolute values of MdPF at baseline cycling did not differ significantly between RT (79.5 ± 8.4 Hz), LL (77.3 ± 7.4 Hz) and AL (82.4 ± 7.9 Hz). When expressed relative to the initial MdPF, the MdPF in LL decreased progressively resulting in a total decrease of 17.2 ± 4.5% at peakP, whereas in RT

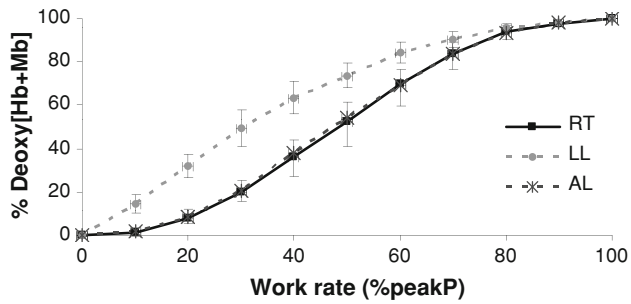

Fig. 2 The mean response of the relative change in deoxy[Hb + Mb] expressed as a function of %peakP for the reference test (RT) and the ramp exercises preceded by a priming leg (LL) and arm exercise (AL)

Table 3 Mean values ± SD for iEMG during baseline cycling, and the slope of the iEMG/*W*-relationship above and below the gas exchange threshold

	iEMG		
	RT	LL	AL
Baseline iEMG (μV)	48.2 ± 8.4	55.4 ± 10.5	52.6 ± 7.2
ΔiEMG/Δ*W* < GET (% W^{-1})	0.74 ± 0.08	0.89 ± 0.11[a,b]	0.72 ± 0.10
ΔiEMG/Δ*W* > GET (% W^{-1})	0.99 ± 0.09	0.93 ± 0.14	1.00 ± 0.15

[a] Significantly different from RT

[b] Significantly different from AL ($P < 0.05$)

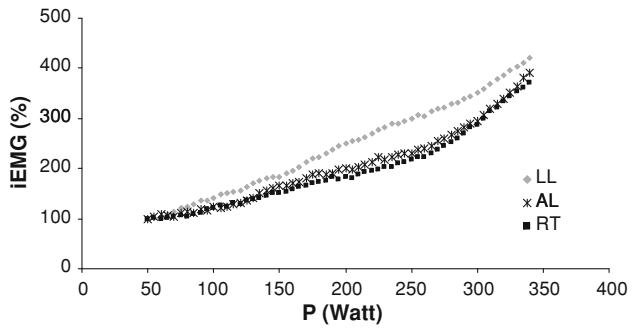

Fig. 3 Integrated EMG (iEMG) as a function of work rate for the reference test (RT) and the ramp exercises preceded by a priming leg (LL) and arm exercise (AL) of a representative subject. Values normalised to those seen at 50 W

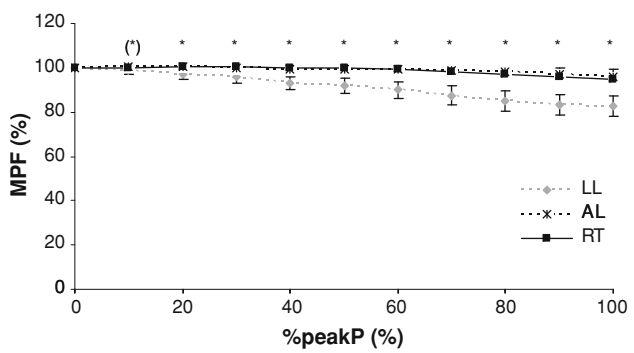

Fig. 4 Median Power Frequency (MdPF) from baseline to peakP (%peakP) for the reference test (RT) and the ramp exercises preceded by a priming leg (LL) and arm exercise (AL). The MdPF is expressed relative to the MdPF at baseline cycling (100%)

(5.0 ± 2.1%) and AL (3.9 ± 3.2%) the decrease in MdPF was significantly less pronounced.

Discussion

In the present study it was found that the deoxy[Hb + Mb] response during incremental ramp exercise was shifted to the left when the exercise was preceded by a high-intensity leg exercise. Instead of the blunted increase (i.e. inherent to the sigmoid pattern) in deoxy[Hb + Mb] during the initial stage of RT and AL, deoxy[Hb + Mb] increased immediately from the onset of ramp exercise when it was preceded by high-intensity leg exercise (LL). This immediate increase in deoxy[Hb + Mb] in LL was accompanied by a higher slope of the $\dot{V}O_2/W$-relationship and iEMG/*W*-relationship below the GET and a more pronounced decrease in MdPF.

Similar to previous studies addressing the pattern of deoxy[Hb + Mb] in response to ramp exercise (Ferreira et al. 2007; Boone et al. 2009, 2010; DiMenna et al. 2010), it was observed in the present study that deoxy[Hb + Mb] increased in a sigmoid fashion as a function of relative work rate. This sigmoid increase has been suggested to reflect the dynamic non-linear relationship between $\dot{Q}_m$ and $\dot{V}O_{2m}$ during this type of non-steady state incremental exercise. More specifically this sigmoid increase indicates the increase in $\dot{V}O_{2m}$ at the onset of the ramp exercise depends to a large part on an increase in $\dot{Q}_m$, rather than on an increase in fractional O_2 extraction. As work rate increases into the low-to-moderate intensity domain, the increase in $\dot{Q}_m$ decelerates slightly so that the increase in $\dot{V}O_{2m}$ will depend progressively on a more pronounced O_2 extraction.

Up to this point, the mechanisms underpinning this sigmoid deoxy[Hb + Mb] response have not been established. In the present study it was investigated whether this typical response is mediated by either systemic or local mechanisms by applying either a priming high-intensity arm or leg exercise. Measurements of lactate concentration were used to quantify the metabolic acidosis which should induce the Bohr effect, i.e. a rightward shift in the HbO_2-dissociation curve. The elevated lactate concentration measured in the present study indicates that this metabolic acidosis was probably present and of a similar magnitude at the onset of the second ramp exercise for both LL and AL. It should be pointed out, however, that measurements of pH would have provided a more accurate indication.

The results of the present study support the hypothesis that the sigmoid increase in deoxy[Hb + Mb] to ramp exercise can mainly be explained at a peripheral level. It

has been shown in rats that muscles consisting of different fibre types are characterised by a specific relationship between O_2 delivery and O_2 extraction. More specifically, the fast-twitch muscles are characterised by a greater fractional O_2 extraction compared to slow-twitch muscles (Behnke et al. 2003; McDonough et al. 2005). In this context, Ferreira et al. (2006) reported different profiles of arteriovenous O_2 difference ($C_{(a\text{-}v)}O_2$)as a function of $\dot{V}O_{2m}$ between 'slow-twitch' and 'fast-twitch' rat muscles. The fast-twitch muscles showed a very steep increase in $C_{(a\text{-}v)}O_2$ even at low metabolic rates, whereas this rapid increase did not occur in the slow-twitch muscles. Since the increase in O_2 delivery for a given change in metabolic rate was similar in the slow-and fast-twitch muscles, the differences in $C_{(a\text{-}v)}O_2$ between the muscles consisting of different fibre types were explained by the lower resting and sub-maximal exercise $\dot{Q}_m$ values in combination with a higher O_2 diffusing capacity in the fast-twitch muscles (i.e. a similar slope of the $\dot{Q}_m/\dot{V}O_{2m}$-relationship with a lower *y*-intercept in the fast-twitch muscles). The higher resting and sub-maximal $\dot{Q}_m$ values could be related to a greater sensitivity to endothelium-dependent vasodilation agents (Wunsch et al. 2000; Woodman et al. 2001) and a lower sympathetically mediated vasoconstriction (Folkow and Halicka 1968; Laughlin and Armstrong 1987; Delp and Armstrong 1988). It can be suggested that at the onset of the ramp exercise predominantly slow-twitch fibres, characterised by a small increase in $C_{(a\text{-}v)}O_2$ are recruited (Krustrup et al. 2004). As work rate increases, the fast-twitch fibres are recruited progressively with a substantially greater increase in $C_{(a\text{-}v)}O_2$ for a similar increase in work rate. This sequential recruitment of muscle fibres, which is typical to incremental exercise, could in this way induce a sigmoid increase in $C_{(a\text{-}v)}O_2$, and thus of deoxy[Hb + Mb]. It can be questioned however, to what extent the observed differences between relatively homogenous "slow-twitch" and "fast-twitch" muscles in rats can apply to changes in muscle fibre recruitment within a heterogenous skeletal muscle like the M. Vastus Lateralis.

The observation of a higher slope of the iEMG/*W*-relationship below the GET in combination with a more pronounced decrease in MdPF in LL compared to RT and AL is in line with the above hypothesis. The shift of the EMG power density spectrum towards lower frequencies has been used as an index of muscle fatigue during both isometric (Hagberg 1981; Masuda et al. 1999) and dynamic exercise (Komi and Tesch 1979, Taylor et al. 1997, Masuda et al. 1999). Therefore, the progressive decrease in MdPF from the onset of LL might reflect the appearance of fatigue in the muscle fibres that were already active in the priming high-intensity leg exercise. Additionally, it has been suggested that priming high-intensity leg exercise alters and/or increases the muscle fibre recruitment of the subsequent exercise (Burnley et al. 2002; Fukuba et al. 2002; Layec et al. 2009). This is in line with the higher slope of the iEMG/W and $\dot{V}O_2/W$-relationship below the GET in the present study. In accordance with the study of Jones and Carter (2004), the higher S1 in LL might be attributed to an altered and/or increased motor unit recruitment. The possible impact of altered muscle fibre recruitment on the $\dot{V}O_2$ response to sub-maximal ramp exercise is based on the different energetic characteristics of slow- and fast-twitch muscle fibres. It has been demonstrated that slow-twitch fibres have a lower O_2 cost (Crow and Kushmerick 1982; Kushmerick et al. 1992) and a lower Ca^{2+}-ATPase activity (Willis and Jackman 1994) for a given force production compared to fast-twitch fibres. Therefore, an increased contribution of fast-twitch fibres in the generation of muscle force might result in a higher slope of the $\dot{V}O_2/W$-relationship below the GET. Since these fast-twitch fibres are characterised by a steep increase in $C_{(a\text{-}v)}O_2$ (Ferreira et al. 2006) and thus deoxy[Hb + Mb] this change in muscle fibre recruitment might contribute to the immediate increase in deoxy[Hb + Mb] and the disturbance of the sigmoid pattern to ramp exercise. The surface EMG shows that this change in muscle fibre recruitment does not occur following prior high-intensity arm exercise and therefore, that the sigmoid pattern of deoxy[Hb + Mb] to the subsequent ramp exercise is not impacted, despite a potential shift in the O_2 dissociation curve due to the establishment of the metabolic acidosis. It could be argued however, that the additional recruitment of motor units per se might induce an increase in S1. It has been reported that 30–50% of the energy required for muscle contraction is related to fibre activation (Szentesi et al. 2001). Therefore, recruitment of additional muscle fibres, next to the fatigued fibres that consume O_2 during recovery but do not generate force might, also result in the higher S1. It cannot be concluded from the present study which of the two mechanisms (or both) is the predominant mechanism underlying the higher S1, or if there is another, not defined, mechanism. Nonetheless, the results of the present study clearly point out the important role of the muscle fibre recruitment and characteristics in the deoxy[Hb + Mb] response as well as the $\dot{V}O_2$ response.

It should be noted however, that the immediate increase in deoxy[Hb + Mb] to ramp exercise following the high-intensity leg exercise might not only be the result of the activity of the fast-twitch fibres at the onset of the ramp exercise. It was observed in the present study that the level of deoxy[Hb + Mb] during baseline cycling tended to be lower during the subsequent ramp exercise (LL) compared to the prior high-intensity leg ramp exercise (RT) (21.7 ± 5.2 vs. 23.6 ± 8.0 μM; $P = 0.065$). This lower

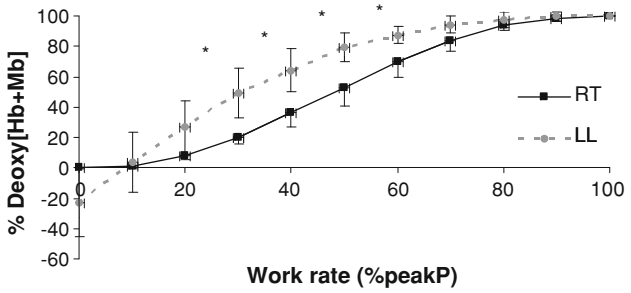

Fig. 5 The mean response of deoxy[Hb + Mb] for the reference test and the ramp exercise preceded by a priming leg exercise (LL) expressed as a function of %peakP. The deoxy[Hb + Mb] response for the LL-test was expressed relative to the total amplitude of the deoxy[Hb + Mb] response of the RT, i.e. considering that the baseline value prior to the onset of the LL-ramp was lower than for the RT

baseline deoxy[Hb + Mb] can be attributed to the hyperaemia following the high-intensity exercise with an increase in O_2 delivery to the muscles (Delorey et al. 2007). Therefore, the response of deoxy[Hb + Mb] at the onset of the subsequent ramp exercise may reflect both ongoing hyperaemia and a change in the recruitment pattern of the muscle fibres. In order to gain insight into the role of this elevated tissue oxygenation (i.e. lower deoxy[Hb + Mb]) resulting from the prior high-intensity leg exercise, we normalised the deoxy[Hb + Mb] response of the subsequent exercise to the end-exercise value of the prior high-intensity exercise test (Fig. 5). It was observed that the baseline deoxy[Hb + Mb] of the subsequent test started approximately 20–25% below the baseline deoxy[Hb + Mb] of the RT. Furthermore, the results of this analysis suggest that the disturbance of the sigmoid pattern of deoxy[Hb + Mb] in the subsequent test, when preceded by high-intensity leg exercise, could be attributed in large part to the relative hyperaemia during recovery which lowered the deoxy[Hb + Mb]. However, the change in deoxy[Hb + Mb] during the subsequent ramp exercise still significantly exceeded the change in deoxy[Hb + Mb] of the RT at 30, 40, 50 and 60% peakP. These differences could be related to a higher recruitment of fast-twitch fibres in the low-to-moderate intensity domain of the subsequent ramp exercise.

It could be argued that the observed differences in the pattern of deoxy[Hb + Mb] to LL compared to RT and AL could also be related to an increase in muscle temperature and skin blood flow due to the priming leg exercise. Davis et al. (2006) and Tew et al. (2010) observed that local heating and also whole body heating induced a significant increase in the oxygenation and saturation of the muscle tissue. However, in the present study it can be expected that an increase in muscle temperature as well as skin blood flow will not have had an important impact on the pattern of deoxy[Hb + Mb] since it has been shown that this parameter is relatively insensitive to changes in blood volume under the NIRS probe (Maehara et al. 1977; Chuang et al. 2002; Grassi et al. 2003; Ferreira et al. 2005). Furthermore, an increased blood flow to the working muscles due to a temperature-mediated vasodilation would only result in a delayed increase in the microvascular O_2 extraction and thus a delay in the increase of deoxy[Hb + Mb] during LL. This is in contrast to the pattern of deoxy[Hb + Mb] in LL observed in the present study and therefore, in our opinion, an increase in temperature and skin blood flow would not have had an important impact on the results of the present study.

In line with studies reporting a reduced amplitude of the $\dot{V}O_2$ slow component during a constant work rate transition following a bout of heavy exercise (Burnley et al. 2000; Fukuba et al. 2002), the slope of the $\dot{V}O_2/W$-relationship above the GET (S2) was reduced in LL compared to RT and AL. The changes in the slow component following heavy exercise were related to alterations in muscle fibre recruitment pattern. Several mechanisms were proposed that might result in a less pronounced recruitment of fast-twitch fibres in the second exercise bout, e.g. a better matching of the O_2 supply to O_2 demand in the microcirculation or an increase in efficiency due to an increase in muscle temperature. However, the results of the present study are not completely in support of these explanations. The immediate increase in deoxy[Hb + Mb] at the onset of LL suggests that these fast-twitch fibres were already active in the low intensity domain and thus that this alteration in the muscle fibre recruitment pattern did not occur during LL. The $\dot{V}O_2/W$-relationship in LL did not show the upward deflection at the GET (S1 and S2 did not differ significantly) as in RT and AL. Although the results of the present study are not in line with the suggested mechanisms for the reduced slow component (Burnley et al. 2000; Fukuba et al. 2002), the results of the present study points out that the slow component and the steeper slope of the $\dot{V}O_2/W$-relationship above the GET are probably related to the muscle fibre recruitment pattern.

Conclusions

From the present study it can be concluded that priming high-intensity leg exercise disturbs the traditionally observed sigmoid pattern of deoxy[Hb + Mb] in response to ramp exercise. When the ramp exercise was preceded by the leg exercise, deoxy[Hb + Mb] increased immediately from the onset of the ramp exercise. The observation that the sigmoid pattern remained unchanged when the exercise was preceded by the high-intensity arm exercise indicates

that the Bohr effect is probably not the mechanism behind the sigmoid increase. It is likely that the mechanisms relate to the relative hyperaemia at the onset of the subsequent ramp exercise and/or to the sequential recruitment of muscle fibre types with different characteristics.

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